

A Review on Polyester Powder Coatings: Materials, Deposition Techniques, Wear Mechanisms, and Nanofiller-Based Performance Enhancement

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ABSTRACT

Powder coating technology has emerged as an environmentally friendly and high-performance surface engineering technique widely adopted in automotive, appliance, construction, and industrial applications. Among various powder coatings, polyester-based systems are extensively used due to their excellent corrosion resistance, mechanical strength, thermal stability, and aesthetic properties. In recent years, the incorporation of nanofillers such as boron carbide (B_4C), silicon dioxide (SiO_2), titanium dioxide (TiO_2), and graphene has attracted significant research attention for enhancing wear resistance, scratch hardness, and durability of polyester powder coatings. This review presents a comprehensive overview of powder coating processes, electrostatic deposition mechanisms, Faraday cage and back-ionization effects, types of thermoplastic and thermosetting powders, and performance evaluation techniques. Special emphasis is placed on tribological behavior, scratch resistance, corrosion performance, and optimization of process parameters using statistical tools such as Taguchi methods. The review identifies key research gaps and highlights future directions for nanocomposite polyester powder coatings in advanced engineering applications.

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1. Introduction

Surface engineering plays a crucial role in enhancing the performance, durability, and service life of engineering components exposed to mechanical wear, corrosion, and harsh environmental conditions [1]. Among various surface protection techniques, powder coating has emerged as a sustainable and high-performance alternative to conventional liquid coating processes due to its solvent-free nature, high material utilization efficiency, superior surface finish, and environmental compliance [2].

Powder coatings are applied in the form of finely divided solid particles that are electrostatically deposited onto a grounded substrate and subsequently cured at elevated temperatures to form a continuous, uniform film. Unlike liquid coatings, powder coatings eliminate volatile organic compounds (VOCs), minimize material wastage, and provide excellent mechanical and chemical resistance. As a result, they are extensively used in automotive, appliance, construction, aerospace, electrical, and industrial equipment applications [3].

Among thermosetting powder coatings, polyester-based coatings have gained widespread industrial acceptance owing to their excellent corrosion resistance, good mechanical strength, thermal stability, weatherability, and aesthetic appeal [4]. Polyester powder coatings are particularly preferred for outdoor applications due to their superior resistance to ultraviolet (UV) radiation and environmental degradation compared to epoxy coatings [5]. However, despite these advantages, conventional polyester coatings often exhibit limited wear resistance, scratch sensitivity, and degradation under severe tribological loading, which restrict their use in high-contact and abrasive environments [6].

To overcome these limitations, significant research efforts have focused on the development of polyester powder coating nanocomposites, where hard ceramic or carbon-based nanofillers such as boron carbide (B_4C), silicon dioxide (SiO_2), titanium dioxide (TiO_2), nanoclay, graphene, and boron nitride are incorporated into the polymer matrix. The addition of nanofillers enhances coating performance by improving hardness, load-bearing capacity, interfacial bonding, wear resistance, and scratch hardness, while maintaining desirable surface finish and corrosion protection [7].

Furthermore, the performance of powder coatings is strongly influenced by process parameters such as particle size, coating thickness, curing temperature, applied load, sliding frequency, and nanofiller content. Therefore, statistical optimization techniques, particularly the Taguchi method and analysis of variance (ANOVA), are widely employed to identify optimal parameter combinations and to achieve enhanced tribological and mechanical performance with minimal experimental effort [8].

This review paper presents a comprehensive overview of powder coating technology, polyester powder coatings, nanofiller reinforcement strategies, tribological behavior, and performance evaluation techniques. Emphasis is placed on wear mechanisms, scratch resistance, corrosion performance, and process optimization approaches [9]. The review also identifies existing research gaps and outlines future directions for the development of high-performance polyester powder coating systems for advanced engineering applications.

2. Literature Review

Polyester powder coatings are widely used for metal protection due to low VOC emissions, good mechanical strength, and corrosion resistance, and their properties can be significantly tailored using micro- and nanofillers [10-23].

2.1 Materials and Base Systems

Thermoset carboxy-functional polyesters are often used alone or in hybrid systems with epoxy (typically 70/30 polyester/epoxy) for improved mechanical and corrosion behavior. Hybrid

polyester/epoxy powder coatings show high corrosion protection (≈ 850 h salt spray) even before nano-optimization. Polyester matrices are also used as binders in functional coatings, including electromagnetic absorbers and corrosion-resistant formulations. Polyesters have inherently good thermal and chemical resistance; synthetic aromatic polyesters provide superior moisture resistance, nonflammability, and mechanical strength compared with aliphatic types, making them ideal matrices for structural and functional nanocomposites.

2. 2 Deposition and Processing of Powder Coatings

Commercial polyester powder coatings are usually deposited by electrostatic spraying onto metal substrates followed by oven curing. For nanofiller incorporation, several dry and wet routes are reported:

- ♦ **Ball milling / planetary milling** of commercial polyester powders with SiO_2 nanoparticles (0.5–3 wt%) yields good dispersion and improved wear resistance with 0.5 wt% silica.
- ♦ **Twin-screw extrusion** is used to incorporate clay minerals (mica, montmorillonite) into hybrid polyester/epoxy powders, giving homogeneous dispersion at 2–6 phr nanofiller.
- ♦ Hot-mixing and extrusion are frequently used in nano-functionalization of epoxy and polyamide powders and are conceptually transferable to polyester systems.
- ♦ Deposition by electrostatic spray is also applied for various polymer nanocomposite coatings (e.g., UHMWPE and PEEK systems), underlining its general suitability for nanopowder-based tribological coatings. Good nanofiller dispersion is critical; agglomerates act as stress concentrators and can degrade mechanical and barrier properties.

2.3 Effect of Conventional Fillers in Polyester Powder Coatings

- ♦ Micro- and submicron fillers are widely used in polyester powders for mechanical tuning: Microsilica ($\sim 2.9 \mu\text{m}$) improves impact strength and bending strength; as average filler size increases, gloss and impact strength decrease, indicating a trade-off between toughness and appearance
- ♦ Wollastonite (fibrous CaSiO_3), due to its acicular crystalline morphology, enhances bending strength and inhibits cracking acting as a reinforcing phase.
- ♦ Clay minerals (muscovite mica, MMT) in hybrid epoxy/polyester powders increase roughness and reduce gloss; mica with lamellar plates yields better mechanical performance than agglomerated MMT. These microfillers provide a baseline understanding of morphology-controlled reinforcement that is further exploited at the nanoscale.

3. Nanofillers in Polyester and Polyester-Based Coatings

Inorganic Oxide and Clay Nanoparticles

Silica, TiO_2 , Al_2O_3 , nanoclay, and layered silicates are among the most studied nanofillers.

1. **SiO_2 nanoparticles** in polyester powders (0.5–3 wt%) increase abrasive and erosion wear resistance; 0.5 wt% gives optimal improvement without compromising appearance. In broader polymer systems, nanosilica enhances hardness, stiffness, and wear resistance in epoxy and polyamide powder coatings, with 1 wt% often being optimal.

2. **Nanoclay (montmorillonite)** in polyester–epoxy powder coatings improves mechanical performance and can enhance corrosion resistance when well exfoliated; poorly dispersed agglomerates lower mechanical properties compared to lamellar mica. In polyester bulk nanocomposites, nanoclay improves mechanical properties and reduces water uptake; optimized contents significantly increase wear resistance as shown by RSM-based pin-on-disc studies. TiO_2 , Al_2O_3 and nanoclay additions to unsaturated polyester and epoxy coatings increase hardness and adhesion and modify surface wettability; 4 wt% TiO_2 in polyester yields the highest hardness among tested combinations.

3.1 Carbon-based nanofillers offer high stiffness, thermal conductivity and barrier effects.

A comprehensive review of polyester nanocomposites with nanodiamond, fullerenes, CNTs, graphene and graphene oxide highlights that optimized functionalization and matrix–filler interaction lead to significant gains in mechanical, thermal, and electrical performance. Graphene-based nanofillers are especially promising for barrier and tribological enhancement. For saturated polyester coatings, graphene nanoplatelets (GNP) and graphene oxide (GO) have been incorporated together with melamine crosslinker (HMMM). Increasing GO or GNP loading tends to reduce glass transition temperature and hardness due to reaction of hydroxyl groups on the nanofiller with HMMM, lowering crosslink density. At fixed loading, GO having more hydroxyl groups more strongly decreases T_g and hardness than GNP. However, increasing HMMM content compensates this effect and can restore or enhance T_g and hardness (e.g., 2 wt% GO nanocomposites show T_g rise from 24.7 °C to 32.1 °C and hardness from 193 to 452 MPa when HMMM/polyester changes from 1:5 to 1:2). (Zhao et al., 2024) This underscores the need to adjust curing chemistry when using highly functionalized nanofillers.

More broadly, carbonaceous nanofillers in polyester matrices have enabled applications in EMI shielding, supercapacitors, fuel cells, textiles, and shape-memory materials, benefiting from tailored electrical conductivity and mechanical strength.

3.2 Other Functional Nanofillers and Hybrid Systems

Nanofillers are also used to impart special functionalities:

1. **Magnetic and dielectric nanoparticles**— NiFe_2O_4 , doped $\text{g-C}_3\text{N}_4$, and MTiO_3 ($M = \text{Fe, Mg, Zn}$)—dispersed in polyester matrices yield high-performance microwave-absorbing coatings. Optimized polyester nanocomposites reach >99% absorption ($\text{RL} < -30$ dB) in X-band with only 1 mm thickness, driven by high attenuation constants (>200 dB/m). (Solgi et al., 2021)-
2. **Multicomponent MXene/PANI/MMT** nanostructures in polyester/epoxy powder coatings drastically enhance corrosion resistance and mechanical properties. 1.5 wt% Ti_3C_2 MXene/PANI/MMT nanosheets in a polyester/epoxy powder yield polarization resistance $\approx 3.39 \times 10^6 \Omega \cdot \text{cm}^2$ and corrosion potential -0.062 V after 42 days NaCl immersion, far better than pristine coatings, while increasing hardness from ≈ 153 to 229 MPa. (Nazarlou et al., 2023)
3. Natural micro-/nanofillers such as **eggshell particles ($\sim 1 \mu\text{m}$)** in unsaturated polyester enhance hardness, flexural strength, and especially wear resistance (up to $\sim 74\%$ improvement at 4 wt% eggshell), while also improving thermal stability and reducing water diffusion. (Jassim & Jassim, 2023)

4. Wear and Tribological Mechanisms in (Polyester) Polymer Coatings

Polymer coatings often fail via abrasive, adhesive, or fatigue wear, which can undermine their corrosion-barrier function. In polyester powder coatings with silica nanoparticles, ball-milled powders yield well-dispersed nanofillers and improved wear performance in pin-on-disc and sand erosion tests. At low nano-SiO₂ contents, improved hardness, stiffness, and possible modification of viscoelastic behavior shift wear from severe abrasive to milder mechanisms; excessive loading can increase brittleness and introduce microcracking. For polyester–nanoclay bulk composites, nanoclay reduces wear rate by improving load-carrying capacity and reducing ploughing, with RSM showing significant contributions from nanoclay content, load, and speed to wear response. Broader tribological reviews of polymer nanocomposites highlight that:

- Optimum nanofiller content is critical; too low gives negligible effect, too high can cause agglomeration and increased friction/wear.
- Nanofillers can promote formation of protective transfer films on the counterface, reduce real contact area, and increase thermal conductivity, all favoring reduced wear rate and coefficient of friction.
- Sliding conditions (load, speed, environment) govern whether nanofillers behave as solid lubricants or hard abrasives. In some systems (e.g., SiO₂/epoxy), high COF results when nanofillers increase shear at contact regions.

Recent work on coatings with multidimensional nanofillers (0D to 3D) emphasizes that particle size, dimensionality and structure strongly influence friction-reduction and wear-resistance mechanisms. One-dimensional and two-dimensional nanofillers (CNTs, graphene) can substantially enhance load-bearing capacity of polymer coatings, addressing the inherent strength limitations of polymers.

4.1 Corrosion and Barrier Performance of Polyester Nanocomposite Coatings

Inorganic nanofillers such as metal oxides, glass flakes, nitrides, carbides, and nano-CaCO₃ in polymer matrices significantly improve corrosion resistance by increasing tortuosity for ion diffusion, filling micro-defects, and enhancing adhesion. (Pourhashem et al., 2020) Key controlling factors include particle type, morphology, size, surface area, filler fraction, chemical structure and functional groups, and interfacial interaction with the polymer. (Pourhashem et al., 2020; Nazarlou et al., 2023; Relosi et al., 2018) Polyester-based powder coatings with montmorillonite nanoparticles on carbon steel demonstrate enhanced corrosion performance due to better barrier action and microstructural densification. (Relosi et al., 2018).

Hybrid polyester/epoxy/MXene-based powders achieve several orders of magnitude higher impedance and polarization resistance compared with unfilled coatings. (Nazarlou et al., 2023)

Review work stresses that well-dispersed nanofillers, sometimes achieved via green processing routes and non-toxic modifications, provide the highest corrosion resistance without damaging intrinsic nanofiller properties. (Pourhashem et al., 2020)

Role of Curing and Crosslinking Chemistry

Nanofillers can significantly affect curing kinetics and network structure of polyester powder coatings:

- Hydroxyl-rich graphene oxide reacts with HMMM crosslinker, reducing effective crosslink density and hence Tg and hardness; adjustment of HMMM content is required to restore properties.
- Nanofillers such as CaCO₃ and TiO₂ have been reported to reduce activation energy of polyester-based resins, modifying curing behavior and potentially influencing final crosslink density and mechanical properties. Understanding filler–matrix
- crosslinker interactions (including hydroxyl, carboxyl, and amine groups) is essential to avoid unintended plasticization or embrittlement while exploiting reinforcement.

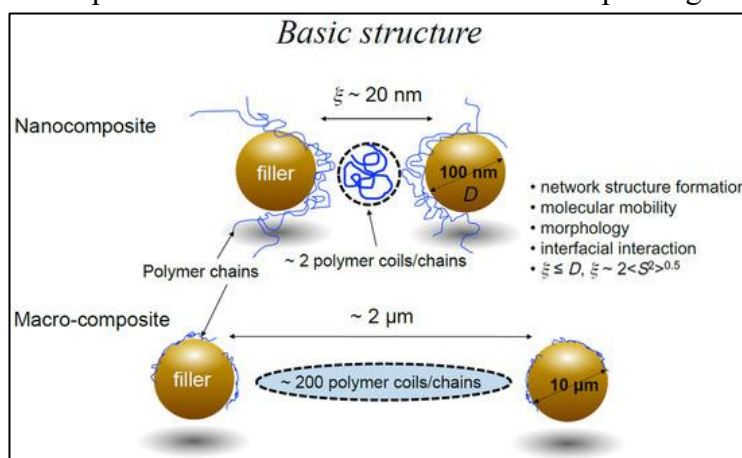


Figure 1. Typical particle reinforcements/geometries in nano- and macro-compo.

5. Synthesis: Design Principles and Research Gaps

Across the literature, several design principles for polyester powder nanocomposite coatings emerge:

- Use **low to moderate nanofiller contents** (typically 0.5–3 wt%) to balance reinforcement against agglomeration and brittleness
- Prioritize **dispersion quality** via ball milling, extrusion, or in situ techniques; poor dispersion compromises mechanical, tribological, and barrier benefits.
- Tailor **curing chemistry and crosslink density** when using reactive nanofillers (GO, nanoclay with surface hydroxyls, conductive polymers) to avoid Tg and hardness losses.
- Match nanofiller morphology and functionality to targeted properties: lamellar or fibrous fillers for crack-bridging, 2D graphene-like fillers for barrier and electrical effects, oxides and clays for hardness and corrosion protection, and functional oxides for microwave absorption. Important research gaps include systematic work specifically on polyester powder (rather than liquid or other polymer) tribological coatings with a wide variety of nanofillers, long-term durability under cyclic mechanical and environmental loading, and industrial-scale dispersion and curing control under realistic production conditions.

Table 1: Key Systems, Properties, and Effects

System / Matrix	Nanofiller type & loading	Main property effects
Polyester powder	SiO ₂ (0.5–3 wt%, ball-milled)	Improved abrasive & erosive wear at 0.5 wt%; good dispersion
Hybrid polyester/epoxy powder	Muscovite mica, MMT (2–6 phr)	Higher roughness, lower gloss, good corrosion protection; mica > MMT in mechanics
Saturated polyester coating	GNP, GO (≤ 5 wt%) + HMMM	GO/GNP lower T _g & hardness via crosslinker consumption; higher HMMM restores/enhances
Unsaturated polyester bulk	Nanoclay (various wt%)	Increased mechanical strength & wear resistance, reduced water uptake
Unsaturated polyester bulk	Eggshell (~ 1 μ m, 4–5 wt%)	Improved hardness, flexural strength, wear resistance, thermal stability
Polyester matrix (functional)	NiFe ₂ O ₄ , doped g-C ₃ N ₄ , MTiO ₃	High microwave absorption (RL < –30 dB, X-band) at 1 mm thickness
Polyester/epoxy powder	Ti ₃ C ₂ MXene/PANI/MMT (1.5 wt%)	Greatly enhanced corrosion resistance and hardness
Polyester (general)	Carbonaceous nanofillers (CNT, graphene, GO, ND, C ₆₀)	Improved mechanical, thermal, electrical, barrier properties; EMI shielding

6. Conclusion

This review has comprehensively examined the development, processing, and performance enhancement of polyester powder coatings, with particular emphasis on the role of micro- and nanofiller reinforcement in improving tribological, mechanical, and corrosion-resistance properties. Polyester powder coatings remain one of the most attractive surface protection technologies due to their environmental compatibility, ease of processing, and excellent balance of mechanical strength, durability, and aesthetics.

The literature clearly demonstrates that incorporation of nanofillers such as SiO₂, TiO₂, Al₂O₃, nanoclays, graphene-based materials, carbon nanotubes, MXenes, and functional oxides can significantly enhance wear resistance, scratch hardness, corrosion protection, and functional performance of polyester-based coatings. However, these improvements are strongly dependent on nanofiller type, morphology, surface chemistry, dispersion quality, and loading level. Most studies report optimal performance at low to moderate nanofiller contents (typically 0.5–3 wt%), while excessive loading often leads to agglomeration, embrittlement, and deterioration of coating properties.

Processing methods such as ball milling, extrusion, and hot mixing, combined with electrostatic powder deposition, are effective routes for producing polyester powder nanocomposite coatings, provided that uniform nanofiller dispersion is achieved. The interaction between nanofillers and curing chemistry plays a critical role in determining final coating performance; reactive nanofillers can alter crosslink density, glass transition temperature, and hardness, necessitating careful optimization of curing agents and processing conditions.

From a tribological perspective, nanofillers improve wear resistance by enhancing load-bearing capacity, modifying viscoelastic behavior, promoting protective transfer films, and reducing real contact area during sliding. In corrosion protection, well-dispersed nanofillers increase barrier efficiency by enhancing diffusion tortuosity and minimizing coating defects. Advanced hybrid systems incorporating multifunctional nanostructures demonstrate exceptional performance, opening new avenues for smart and functional powder coatings.

Despite these advances, significant research gaps remain. Systematic studies focused specifically on polyester powder coatings—as distinct from liquid or bulk polymer systems—are limited, particularly with respect to long-term durability under cyclic mechanical, thermal, and environmental loading. Additionally, challenges related to industrial-scale nanofiller dispersion, curing control, and cost-effectiveness require further investigation.

Overall, polyester powder nanocomposite coatings represent a promising class of advanced surface engineering materials. Future research integrating optimized nanofiller design, advanced processing routes, statistical optimization, and real-world performance validation will be essential to fully exploit their potential in high-demand industrial applications.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this manuscript.

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